

STAT3003: Survey Methods

3. Stratified Random Sampling

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3.1 Introduction

- In the previous lecture we covered our first survey design: **simple random sampling**
- We discussed
 - How to **perform it**
 - What the **best estimators** for population mean, total and proportion are, under this design
 - **Properties** of these estimators (bias, variance etc.)
 - How to **determine sample size**

3.1 Introduction

- We will address the same issues this questions this lecture, but for our second sample survey design...
- Stratified Random Sampling

3.1 Introduction

- You wish to estimate the average wage of workers in a city
- The city has a population of 9, with three companies (A, B and C) employing 3 people each
- The breakdown of monthly wages by company is
 - Company A: \$100,000, \$105,000, \$97,5000
 - Company B: \$50,000, \$47,800, \$49,250
 - Company C: \$12,000, \$11,500, \$10,900

3.1 Introduction

- You have two sampling methods available:
 - A **simple random sample**: randomly select 3 people from the population of 9, take the average
 - A **stratified random sample**: randomly select 1 person from each company, take the average
- Which do you expect will produce the closer estimate?

3.1 Introduction

- Why this difference?
- Because the **variance within** each Company is small, but the **variance between** them is large
- Roughly speaking, we shall see that the variance of the estimator under stratified random sampling **depends only on the variance within each stratum**
- Whereas the variance of the estimator under SRS **depends on the population variance**, which includes the variance within strata and variance between strata

3.1 Introduction

- Definition
 - A stratified random sample is one obtained by **partitioning** the population elements into groups called *strata*, then **performing SRS within** each *stratum*
- Thus performing a stratified random sample consists of two parts
 1. Choosing **non-overlapping** strata for your population
 2. Performing multiple, **independent** SRSs

3.1 Introduction

- What's the point?
 - **Reduction of estimator variance**: an estimator under stratified random sampling can have smaller variance (and therefore tighter confidence intervals)
 - **Reduction of cost**: the cost per observation may be reduced if the strata are convenient
 - **Comparison among sub-groups**: the simple random samples among the strata are independent, enabling comparison of estimates
 - **Prevent really bad (unrepresentative) samples**: sampling from strata ensures elements from those strata will be represented in the sample

3.1 Introduction

- Choosing the strata can be difficult and counter-productive – see the “Cautionary Example” at the end of this lecture
- The criteria need to be **specific** so the **strata do not overlap**
- A **poor choice** of strata can lead to the variance of the estimator being larger than that of a simple random sample
- Once the strata have been chosen, you need to **decide on the sample size in each**. There are many ways of doing this (we shall see later)

3.1 Introduction

- It is very important that the simple random samples conducted within each stratum are **independent**
- So **different random sampling schemes** (lots, random numbers etc.) should be used for each stratum
- So that observations in one stratum do not depend on those in another
- This allows later **comparison** of estimates between the strata

3.1 Introduction

- Example
 - You want to conduct a survey to find out how many hours a week the households in your local area watch TV.
 - Your local area contains a public housing estate, a collection of village houses and a new development aimed at young professionals
 - Why might you consider using stratified sampling for your survey?

3.1 Introduction

- Example
 - You wish to conduct a study to investigate the total of worker-hours lost every year due to accidents among all CUHK staff
 - CUHK staff composes of professors, lab technicians, labourers, administrators, monkey-catchers etc.
 - Why might you consider using stratified random sampling for your survey?

3.2 Estimating τ , μ and p

- Before we crack on with defining the estimators and finding their properties, we should define some notation
 - L = Number of strata
 - N_i = Number of sampling units with stratum i
 - N = Number of sampling units in population, so $N = N_1 + N_2 + \dots + N_L$
 - $\bar{Y}_{st}$ = Estimator of population mean from stratified sampling
 - A subscript of i will mean that the parameter/estimator concerned is from stratum i . E.g. $\bar{Y}_i, \mu_i, s_i^2, \tau_i$ etc.

3.2 Estimating τ , μ and p

- This time we start by considering the estimation of the population total, τ
- We have seen previously that $N_i \bar{Y}_i$ is an unbiased estimator of population total τ_i
- Clearly $\tau = \tau_1 + \tau_2 + \cdots + \tau_L$ so $\sum_{i=1}^L N_i \bar{Y}_i$ will be our estimator for τ
- $E\left(\sum_{i=1}^L N_i \bar{Y}_i\right) = \sum_{i=1}^L E(N_i \bar{Y}_i) = \sum_{i=1}^L \tau_i = \tau$, so our estimator is unbiased.
- Let us denote $\sum_{i=1}^L N_i \bar{Y}_i$ by $\hat{\tau}_{st}$

3.2 Estimating τ , μ and p

- How about the variance?
 - $Var(\hat{\tau}_{st}) = Var(\sum_{i=1}^L N_i \bar{Y}_i) = \sum_{i=1}^L Var(N_i \bar{Y}_i)$
because the $\bar{Y}_i$ are independent
 - $\sum_{i=1}^L Var(N_i \bar{Y}_i) = \sum_{i=1}^L N_i^2 Var(\bar{Y}_i)$
 - $= \sum_{i=1}^L N_i^2 \frac{\sigma_i^2}{n_i} \left(\frac{N_i - n_i}{N_i - 1} \right)$ using the formula from SRS
- Notice that the variance of the estimator depends on σ_i^2 's and not σ^2

3.2 Estimating τ , μ and p

- Again, the variance of our estimator contains parameters we usually do not know, the σ_i^2 's
- Thus we need an unbiased estimator for the variance of our estimator
- Luckily, from our work on simple random sampling, we know an unbiased estimator for

$$\sigma_i^2: \left(1 - \frac{1}{N_i}\right) \hat{\sigma}_i^2$$

3.2 Estimating τ , μ and p

- Plugging our unbiased estimator for σ_i^2 into our formula for $Var(\hat{\tau}_{st})$, we arrive at an unbiased estimator for $Var(\hat{\tau}_{st})$:

- $\widehat{Var}(\hat{\tau}_{st}) := \sum_{i=1}^L N_i^2 \frac{1}{n_i} \left(\frac{N_i - n_i}{N_i - 1} \right) \left(1 - \frac{1}{N_i} \right) \hat{\sigma}_i^2$
- $= \sum_{i=1}^L N_i^2 \left(1 - \frac{n_i}{N_i} \right) \frac{\hat{\sigma}_i^2}{n_i}$

3.2 Estimating τ , μ and p

- Now we know $\widehat{Var}(\hat{\tau}_{st})$, we can create confidence intervals for τ
- If all the stratum sample size are **small**, an approximate $100(1 - \alpha)\%$ C.I. for τ
 - $\hat{\tau}_{st} \pm t_{df, 1 - \frac{\alpha}{2}} \sqrt{\widehat{Var}(\hat{\tau}_{st})}$
- The question is, what should we use for the df ?
 - Intuitively, we might guess $df = df_1 + df_2 + \dots + df_L$ where $df_k = n_k - 1$, the df from the SRS in stratum k
 - But this is correct only **if the variances of all the strata are the same**

3.2 Estimating τ , μ and p

- Say we have a linear combination of sample variances, χ'
 - $\chi' = \sum_{i=1}^n k_i \hat{\sigma}_i^2$
- We don't know the distribution of χ' . However, we can approximate it well by a chi-squared distribution whose d.f. is given by
 - $df_{\chi'} \approx \frac{(\sum_{i=1}^n k_i \hat{\sigma}_i^2)^2}{\sum_{i=1}^n \frac{(k_i \hat{\sigma}_i^2)^2}{n_i - 1}}$, where $\hat{\sigma}_i^2$ has sample size n_i
 - This is called **Satterthwaite's Approximation**

3.2 Estimating τ , μ and p

- Thus, in our case,
 - $df \approx \frac{(\sum_{h=1}^L k_h s_h^2)^2}{\sum_{h=1}^L \frac{(k_h s_h^2)^2}{n_h - 1}}$ where $k_h = \frac{N_h(N_h - n_h)}{n_h}$
 - Calculating df will produce a non-integer. Be conservative and round it down
 - Thompson (2012) advocates a rule of thumb:
 - If the strata sample sizes are at least 30, we can avoid calculating df by using the normal distribution
 - $\hat{\tau}_{st} \pm z_{1-\frac{\alpha}{2}} \sqrt{\widehat{Var}(\hat{\tau}_{st})}$

3.2 Estimating τ , μ and p

- Finally, we want to find an appropriate sample size $n = n_1 + n_2 + \dots + n_L$
 - given our desired width of C.I. $2d$
 - Or given our desired bound on the error of estimation, d
- As before, this requires us solving
 - $z_{1-\frac{\alpha}{2}}\sqrt{Var(\hat{t}_{st})} = d$ for n
- But we need to know
 - The variances in each strata, i.e. σ_i
 - the relationship between each n_i and n
- All we know is that n_i is some fraction of n
 - So let us write $n_i = na_i$ for $i = 1, \dots, L$

3.2 Estimating τ , μ and p

- Solving for n , we reach

$$\bullet \quad n = \frac{\sum_{i=1}^L N_i^3 \frac{1}{a_i} \frac{\sigma_i^2}{N_i - 1}}{\frac{d^2}{z_{1-\frac{\alpha}{2}}^2} + \sum_{i=1}^L N_i^2 \frac{\sigma_i^2}{N_i - 1}}$$

- As before, we usually don't know the σ_i s, but we may have reasonable guesses v_i s
 - So plug in the v_i s in place of the σ_i s
- Think about trying to find n if we replaced $z_{1-\frac{\alpha}{2}}$ with $t_{df, 1-\frac{\alpha}{2}} \dots$
 - Can we do it?

3.2 Estimating τ , μ and p

- A politician is thinking of running a TV advert attacking her opponent in a small town.
- She wishes to estimate the total hours of TV watched in the town.
- The town consists of parts A, B and C
 - Part A surrounds a factory. Most households are families with parents working in the factory.
 - Most households in Part B are retired couples.
 - Part C is rural, with most households belonging to farming families

3.2 Estimating τ , μ and p

- The politician's team of statisticians decide to use stratified random sampling to estimate τ , the total hours each household will watch TV each week
- The summary statistics are

Part	N_i	n_i	$\bar{y}_i$	s_i
A	155	20	33.90	5.95
B	62	8	25.12	15.25
C	93	12	19.00	9.36

- Find a 95% C.I. for τ

3.2 Estimating τ , μ and p

- We now move on to estimating the population mean, μ
- Given that $\tau = N\mu$, we can find the properties of our estimator for μ quite easily
- We define our estimator for μ , denoted $\bar{Y}_{st}$, by remembering the fact that $\mu = \frac{\tau}{N}$
 - $\bar{Y}_{st} := \frac{1}{N} \hat{\tau}_{st} = \frac{1}{N} \sum_{i=1}^L N_i \bar{Y}_i$

3.2 Estimating τ , μ and p

- Finding the variance of $\bar{Y}_{st}$ is easy:
 - $Var(\bar{Y}_{st}) = Var\left(\frac{1}{N} \hat{t}_{st}\right) = \frac{1}{N^2} Var(\hat{t}_{st})$
 - So $Var(\bar{Y}_{st}) = \frac{1}{N^2} \sum_{i=1}^L N_i^2 \frac{\sigma_i^2}{n_i} \left(\frac{N_i - n_i}{N_i - 1}\right)$
- And so is finding an unbiased estimator for $Var(\bar{Y}_{st})$
 - We replace σ_i^2 's by their unbiased estimators:
 - $\widehat{Var}(\bar{Y}_{st}) = \frac{1}{N^2} \sum_{i=1}^L N_i^2 \left(1 - \frac{n_i}{N_i}\right) \frac{\hat{\sigma}_i^2}{n_i}$

3.2 Estimating τ , μ and p

- The C.I.s for μ quickly follow
- If the sample size in each strata is bigger than 30, we may use

- $\bar{y}_{st} \pm z_{1-\frac{\alpha}{2}} \sqrt{\widehat{Var}(\bar{y}_{st})}$

- If any of the sample sizes are small, we use

- $\bar{y}_{st} \pm t_{df, 1-\frac{\alpha}{2}} \sqrt{\widehat{Var}(\bar{y}_{st})}$

- With $df \approx \frac{(\sum_{h=1}^L k_h s_h^2)^2}{\sum_{h=1}^L \frac{(k_h s_h^2)^2}{n_h - 1}}$ where $k_h = \frac{N_h(N_h - n_h)}{\textcolor{red}{N}^2 n_h}$

3.2 Estimating τ , μ and p

- To find an appropriate sample size n to achieve our given C.I. width d , we again write the sample size in each stratum as a fraction of n : $n_i := a_i n$
- Then we solve $z_{1-\frac{\alpha}{2}} \sqrt{\text{Var}(\bar{Y}_{st})} = d$ for n , which yields

$$n = \frac{\sum_{i=1}^L N_i^3 \frac{1}{a_i} \frac{\sigma_i^2}{N_i - 1}}{\frac{N^2 d^2}{z_{1-\frac{\alpha}{2}}^2} + \sum_{i=1}^L N_i^2 \frac{\sigma_i^2}{N_i - 1}}$$

3.2 Estimating τ , μ and p

- As before, we may treat estimating p as estimating the total number of people in a population who have a particular characteristic
- Given a simple random sample, we saw that $N_i\hat{p}_i$ was a good estimator of N_ip_i the number of people in a population of size N_i who have a particular characteristic.
- Thus, in stratified random sample,
 $N_1\hat{p}_1 + N_2\hat{p}_2 + \cdots + N_L\hat{p}_L$ would be a good estimator of the number of people in a population of size N who have a particular characteristic

3.2 Estimating τ , μ and p

- Thus we define $\hat{p}_{st}$ as our estimator of the population proportion p , where
 - $\hat{p}_{st} = \frac{1}{N} \sum_{i=1}^L N_i \hat{p}_i$
- The variance of $\hat{p}_{st}$ is easy to find, remembering the variance of $\hat{p}_i$ under SRS:
 - $Var(\hat{p}_{st}) = \frac{1}{N^2} \sum_{i=1}^L N_i^2 Var(\hat{p}_i)$
 - $= \frac{1}{N^2} \sum_{i=1}^L N_i^2 \frac{N_i - n_i}{N_i - 1} \frac{1}{n_i} p_i (1 - p_i)$

3.2 Estimating τ , μ and p

- Like before, $Var(\hat{p}_{st})$ contains parameters we do not know. We need an unbiased estimator for it
- $Var(\hat{p}_{st})$ is a linear sum of population variances for each stratum i , σ_i^2 , where the population consists of N_i zeros and ones
- In such populations, $\sigma_i^2 = p_i(1 - p_i)$
- Recall that an unbiased estimator for σ_i^2 is $\left(1 - \frac{1}{N_i}\right) \hat{\sigma}_i^2$

3.2 Estimating τ , μ and p

- We have also shown that, when the population consists of zeros and ones, the sample variance can be calculated by $\hat{\sigma}_i^2 = \frac{n_i}{n_i-1} \hat{p}_i(1 - \hat{p}_i)$
- So by replacing $p_i(1 - p_i)$ in our formula for $Var(\hat{p}_{st})$ by $\left(1 - \frac{1}{N_i}\right) \frac{n_i}{n_i-1} \hat{p}_i(1 - \hat{p}_i)$ we find an unbiased estimator for $Var(\hat{p}_{st})$:
 - $\widehat{Var}(\hat{p}_{st}) = \frac{1}{N^2} \sum_{i=1}^L N_i^2 \left(1 - \frac{n_i}{N_i}\right) \frac{1}{n_i-1} \hat{p}_i(1 - \hat{p}_i)$

3.2 Estimating τ , μ and p

- The C.I.s for p quickly follow
- If the sample size in each strata is bigger than 30, we may use

- $\hat{p}_{st} \pm z_{1-\frac{\alpha}{2}} \sqrt{\widehat{Var}(\hat{p}_{st})}$

- If any of the sample sizes are small, we use

- $\hat{p}_{st} \pm t_{df, 1-\frac{\alpha}{2}} \sqrt{\widehat{Var}(\hat{p}_{st})}$

- With $df \approx \frac{(\sum_{h=1}^L k_h s_h^2)^2}{\sum_{h=1}^L \frac{(k_h s_h^2)^2}{n_h - 1}}$ where $k_h = \frac{N_h(N_h - n_h)}{\textcolor{red}{N}^2 n_h}$

3.2 Estimating τ , μ and p

- Finally, we want to find an appropriate sample size $n = n_1 + n_2 + \cdots + n_L$
 - given our desired width of C.I. $2d$
 - Or given our desired bound on the error of estimation, d
- As before, this requires us solving
 - $z_{1-\frac{\alpha}{2}}\sqrt{Var(\hat{p}_{st})} = d$ for n
- But we need to know
 - The true proportion in each stratum, p_i
 - Our chosen allocation, e.g. the a_i s.t. $n_i = na_i$

3.2 Estimating τ , μ and p

- Solving $d = z_{1-\frac{\alpha}{2}} \sqrt{Var(\hat{p})}$ for n yields
 - $$n = \frac{\sum_{i=1}^L N_i^3 \frac{1}{a_i} \frac{p_i(1-p_i)}{N_i-1}}{\frac{N^2 d^2}{z_{1-\frac{\alpha}{2}}^2} + \sum_{i=1}^L N_i^2 \frac{p_i(1-p_i)}{N_i-1}}$$
- Of course, we don't know the p_i s – that's the purpose of the survey after all, but we may have reasonable guesses v_i s
 - So plug in the v_i s in place of the p_i s
- Alternatively, if we have no idea what p_i is likely to be, plug in $p_i = 0.5$

3.2 Estimating τ , μ and p

- This complicated formula rapidly reduces to something simple if we make some reasonable assumptions
- Assume $\frac{N_i}{N_i-1} \approx 1$
 - $$n = \frac{\sum_{i=1}^L N_i^2 \frac{1}{a_i} p_i(1-p_i)}{\frac{N^2 d^2}{z^2 \frac{1-\alpha}{1-\frac{\alpha}{2}}} + \sum_{i=1}^L N_i p_i(1-p_i)}$$
- Take $p_i = 0.5$
 - $$n = \frac{\sum_{i=1}^L N_i^2 \frac{1}{a_i}}{\frac{4N^2 d^2}{z^2 \frac{1-\alpha}{1-\frac{\alpha}{2}}} + N}$$
- Use proportional allocation
 - $$n = \frac{N^2}{\frac{4N^2 d^2}{z^2 \frac{1-\alpha}{1-\frac{\alpha}{2}}} + N}$$
- Use a 95% C.I.
 - $$n = \frac{N}{1.04 d^2 N + 1}$$
- Say N is very large
 - $$n = \frac{1}{1.04 d^2}$$

3.3 Sample Allocation

- How should we decide these a_i (equivalently, the n_i , the sample size in the i th stratum)?
- The n_i also appear in the variance of our estimator:
 - $$Var(\hat{t}_{st}) = \sum_{i=1}^L N_i^2 \frac{\sigma_i^2}{n_i} \left(\frac{N_i - n_i}{N_i - 1} \right)$$
 - $$Var(\bar{Y}_{st}) = \frac{1}{N^2} \sum_{i=1}^L N_i^2 \frac{\sigma_i^2}{n_i} \left(\frac{N_i - n_i}{N_i - 1} \right)$$
- Our objective is the following: to obtain a specified amount of information at a minimum cost

3.3 Sample Allocation

- Thus the best allocation scheme will be affected by
 - ?
 - ?
 - ?

3.3 Sample Allocation

- Thus the best allocation scheme $a_1, \dots, a_L$ or $n_1, \dots, n_L$ will be affected by
 - The “population” size of each stratum (a sample of 10 from a population of 20 contains more information than a sample of 10 out of 2000000)
 - The variability of observations within the stratum (the higher the variability (σ_i), the larger the sample needed to obtain a good estimate)
 - The cost of obtaining observations from each stratum (the cost may vary from stratum to stratum)

3.3 Sample Allocation

- Choose sample sizes $n_1, \dots, n_L$ that are in the same proportion to the total sample size n as the $N_1, \dots, N_L$ are to the total population size N
- Therefore $n_1 = \frac{nN_1}{N}, n_2 = \frac{nN_2}{N}, \dots, n_L = \frac{nN_L}{N}$
 - And $a_1 = \frac{N_1}{N}, a_2 = \frac{N_2}{N}, \dots, a_L = \frac{N_L}{N}$
- This is simplistic – it does not consider the variability within each stratum
 - If we know about the variances inside each stratum, we can allocate our sample better

3.3 Sample Allocation

- Assume the following
 - You have **no overhead cost**
 - The **cost of collecting data is the same** in each stratum
 - Your **total sample size is fixed at n**
- Then how should we choose the a_i (or n_i)?
- We choose the a_i that **minimize the variance of our estimator**, $Var(\bar{Y}_{st})$

3.3 Sample Allocation

- Mathematically, this means
 - minimizing the function $\frac{1}{N^2} \sum_{i=1}^L N_i^2 \frac{\sigma_i^2}{n_i} \left(\frac{N_i - n_i}{N_i - 1} \right)$
 - over all $n_1, n_2, \dots, n_L$
 - with the constraint $n_1 + n_2 + \dots + n_L = n$
 - This is called Neyman allocation
- This is an example of optimization with constraints
 - The technique to solve such problems uses Lagrange Multipliers

3.3 Sample Allocation

- Consider the problem
 - Maximize/minimize $f(x_1, \dots, x_n)$
 - Subject to the constraint $g(x_1, \dots, x_n) = 0$
- We solve this by
 1. Writing down the **Lagrangian**
$$L(x_1, \dots, x_n, \lambda) = f(x_1, \dots, x_n) - \lambda g(x_1, \dots, x_n)$$
 2. Find the **stationary points** of $L(x_1, \dots, x_n, \lambda)$, i.e. where $\frac{\partial L}{\partial x_i} = 0$ for $i = 1, \dots, n$ and $\frac{\partial L}{\partial \lambda} = 0$
 3. Evaluate $f(x_1, \dots, x_n)$ at each stationary point, **determine which is the maximum/minimum**

3.3 Sample Allocation

- Example
 - Find the maximum and minimum of
$$f(x, y, z) = x + y + z$$
 - subject to the constraint $x^2 + y^2 + z^2 = 1$
- Solution
 - Write down the Lagrangian:
 - $L(x, y, z, \lambda) = x + y + z - \lambda(x^2 + y^2 + z^2 - 1)$
 - Find the stationary points. First find the partial derivatives:
 - $\frac{\partial L}{\partial x} = 1 - 2\lambda x, \frac{\partial L}{\partial y} = 1 - 2\lambda y, \frac{\partial L}{\partial z} = 1 - 2\lambda z,$
$$\frac{\partial L}{\partial \lambda} = -(x^2 + y^2 + z^2 - 1)$$

3.3 Sample Allocation

– Set them to zero and create a system of equations

- $\lambda x = \frac{1}{2}$
- $\lambda y = \frac{1}{2}$
- $\lambda z = \frac{1}{2}$
- $x^2 + y^2 + z^2 = 1$

– Now solve them

- $\Rightarrow \lambda^2(x^2 + y^2 + z^2) = \lambda^2 = \frac{3}{4}$
- $\Rightarrow$ if $\lambda = -\frac{\sqrt{3}}{2}$, $(x, y, z) = \left(-\frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}}\right)$
- $\Rightarrow$ if $\lambda = \frac{\sqrt{3}}{2}$, $(x, y, z) = \left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right)$

3.3 Sample Allocation

– Now evaluate $f(x, y, z)$ at these points

- $f\left(-\frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}}\right) = -\sqrt{3}$

- $f\left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right) = \sqrt{3}$

– Conclude

- The constrained maximum value of $\sqrt{3}$ occurs when

$$x = y = z = \frac{1}{\sqrt{3}}$$

- The constrained minimum value of $-\sqrt{3}$ occurs when

$$x = y = z = -\frac{1}{\sqrt{3}}$$

3.3 Sample Allocation

- Back to our problem

- minimize the function

$$f(n_1, \dots, n_L) = \frac{1}{N^2} \sum_{i=1}^L N_i^2 \frac{\sigma_i^2}{n_i} \left(\frac{N_i - n_i}{N_i - 1} \right)$$

- over all $n_1, n_2, \dots, n_L$

- with the constraint

$$g(n_1, \dots, n_L) = n - \sum_{i=1}^L n_i = 0$$

- Let's solve it

- Write down the Lagrangian

- $L(n_1, \dots, n_L, \lambda) = \frac{1}{N^2} \sum_{i=1}^L N_i^2 \frac{\sigma_i^2}{n_i} \left(\frac{N_i - n_i}{N_i - 1} \right) - \lambda(n - \sum_{i=1}^L n_i)$

3.3 Sample Allocation

– Find the partial derivatives

- $\frac{\partial L}{\partial n_i} = -\frac{N_i^3}{N_i-1} \frac{1}{N^2} \frac{\sigma_i^2}{n_i^2} + \lambda$ for $i = 1, \dots, L$ (exercise!)
- $\frac{\partial L}{\partial \lambda} = (\sum_{i=1}^L n_i - n)$

– Set them to zero and create a system of equations

- $\lambda n_i^2 = \frac{N_i^3}{N_i-1} \frac{1}{N^2} \sigma_i^2 \Rightarrow n_i = \frac{1}{\sqrt{\lambda}} \sqrt{\frac{N_i}{N_i-1} \frac{N_i}{N}} \sigma_i$ for $i = 1, \dots, L$
- $\sum_{i=1}^L n_i = n \Rightarrow \frac{1}{\sqrt{\lambda}} = \frac{n}{\sum_{i=1}^L \sqrt{\frac{N_i}{N_i-1} \frac{N_i}{N}} \sigma_i}$

– And solve

- $n_i = n \frac{\sqrt{\frac{N_i}{N_i-1}} N_i \sigma_i}{\sum_{i=1}^L \sqrt{\frac{N_i}{N_i-1}} N_i \sigma_i}$
- This is the solution of Neyman allocation.
- Trust me, these n_i minimize $Var(\bar{Y}_{st})$

3.3 Sample Allocation

- Neyman allocation is an example of optimal allocation
 - We can optimise over other factors as well
- Assume the following
 - You have a **fixed cost, c_0**
 - The cost of obtaining a single observation from stratum i is **c_i** , $i = 1, \dots, L$
 - Your total budget is **c**
- Then you choose the a_i that minimize $V(\bar{Y}_{st})$ subject to the constraint that
 - **$c_0 + c_1n_1 + c_2n_2 + \dots + c_Ln_L = c$**

3.3 Sample Allocation

- The solution to this (and a question in Problem Sheet 2) is

- $$n_i = \frac{(c - c_0) \left(\sqrt{\frac{N_i}{N_i - 1}} \frac{N_i \sigma_i}{\sqrt{c_i}} \right)}{\sum_{i=1}^L \sqrt{\frac{N_i}{N_i - 1}} N_i \sigma_i \sqrt{c_i}}$$

- As the cost of sampling in a stratum increases, the optimal sample size in that stratum decreases
- If c_i are the same for $i = 1, \dots, L$, this becomes the same solution as Neyman allocation (exercise)

3.3 Sample Allocation

- **A Cautionary Example** : Is it always true that stratified random sampling produces an estimator with smaller variance than SRS?
 - A food distributor wishes to add a new product to the restaurants he supplies
 - So he adds his new product to a sample of restaurants in Hong Kong
 - He supplies 4 major chains: Maxim ($N_1 = 24$ stores), Café de Coral ($N_2 = 36$), Fairwood ($N_3 = 30$) and Yoshinoya ($N_4 = 30$)
 - He wants to estimate μ , the mean sales of the new product per store per month.
 - The distributor only has enough resources to obtain data from $n = 20$ stores

3.3 Sample Allocation

- He decided to use stratified sampling, with each chain acting as a stratum
- As he has no idea what the σ_i^2 are, and the cost of sampling is the same in each stratum, he uses proportional allocation
 - He samples $n_1 = \frac{24}{120} \times 20 = 4$ restaurants from Maxim, $n_2 = \frac{36}{120} \times 20 = 6$ restaurants from Café de Coral, etc.

3.3 Sample Allocation

- He observes the following data of sales in the month
 - Work out $\bar{y}_{st}$, his estimate for the average sales for the month and provide a 95% C.I. for μ

Maxim	Café de Coral	Fairwood	Yoshinoya
94	91	108	92
90	99	96	110
102	93	100	94
110	105	93	91
	111	93	113
	101		
$\hat{y}_1 = 99,$ $s_1^2 = 78.67$	$\hat{y}_1 = 100,$ $s_1^2 = 55.60$	$\hat{y}_1 = 98,$ $s_1^2 = 39.50$	$\hat{y}_1 = 100,$ $s_1^2 = 112.50$

3.3 Sample Allocation

- Now imagine that the distributor had used SRS instead, with exactly the same sample of 20 restaurants (the same data)
 - Work out $\bar{y}$, and calculate the 95% C.I.
 - Compare your answers with those from the stratified sample – what do you notice? What do you think has happened? What can be done to correct this?

3.3 Sample Allocation

- The lesson is
 - Stratifying will not necessarily result in a smaller variance
 - If your reason for stratifying is to have a smaller error of estimation, choose strata whose observations within are likely to be homogeneous
- You may have other reasons for stratifying
 - Convenience
 - You want to estimate parameters for subgroups of the population